

Math Class 11 Federal Board

Solved Test of Chapter 4

(Sequence and Series) and

Chapter 6 (Permutation and

Combination)

For an arithmetic-geometric series:

$$1 + 5\left(\frac{1}{2}\right)^1 + 9\left(\frac{1}{2}\right)^2 + 13\left(\frac{1}{2}\right)^3 + \dots$$

- (a) Find general term of the series.
(b) Sum the series upto n terms,
(c) Sum the series upto infinite terms.

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A.P: 1, 5, 9, 13, ..

$a = 1, d = 4$

G.P: $\left(\frac{1}{2}\right)^0, \left(\frac{1}{2}\right)^1, \left(\frac{1}{2}\right)^2, \dots$

$r = \frac{1}{2}$

a) $T_n = [a + (n-1)d] r^{n-1}$
 $T_n = [1 + (n-1)4] \left(\frac{1}{2}\right)^{n-1}$
 $T_n = (4n-3) \left(\frac{1}{2}\right)^{n-1}$ ✓

b) $S_n = \frac{a}{1-r} + \frac{dr(1-r^n)}{(1-r)^2} - \frac{(a+nd)r^n}{1-r}$

$S_n = \frac{1}{1-\frac{1}{2}} + \frac{4 \cdot \frac{1}{2} (1 - (\frac{1}{2})^n)}{(1-\frac{1}{2})^2} - \frac{(1+n \cdot 4) (\frac{1}{2})^n}{1-\frac{1}{2}}$
 $= 2 + [2 - 2 \cdot (\frac{1}{2})^n] 4 - 2 - 8n (\frac{1}{2})^n$
 $= 2 + 8 - 8 (\frac{1}{2})^n - 2 (\frac{1}{2})^n - 8n (\frac{1}{2})^n$
 $= 10 - (8 + 2 + 8n) (\frac{1}{2})^n$

Find sum of the series

(a) $\sum_{i=1}^n \frac{i}{7^i}$ and (b) $\sum_{i=1}^{\infty} \frac{i}{7^i}$

$S_n = 10 - (10 + 8n) \left(\frac{1}{2}\right)^n$

$S_{\infty} = \frac{a}{1-r} + \frac{dr}{(1-r)^2} = \frac{1}{1-\frac{1}{2}} + \frac{4 \cdot \frac{1}{2}}{(1-\frac{1}{2})^2}$

$S_{\infty} = 2 + 2 = 2 + 8 \Rightarrow S_{\infty} = 10$

$$S_{\infty} = 2 + \frac{2}{1/4} = 2 + 8 \Rightarrow \boxed{S_{\infty} = 10}$$

$$\binom{n}{0} = 1, \binom{n}{1} = n$$

A company has 7 engineers and 5 managers. In how many ways can a project team of 5 be selected if it includes:

- (a) 3 engineers and 2 managers?
- (b) 4 engineers and 1 manager?
- (c) At least 4 engineers?
- (d) At the most 1 manager?

$$\binom{n}{r} = \binom{n}{n-r}$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$n \geq r$$

$$r \leq n$$

$$a) \binom{7}{3} \binom{5}{2} = \frac{7!}{3! \cdot 4!} \times \frac{5!}{2! \cdot 3!}$$

$$= 35 \times 10 = 350$$

$$\frac{7 \cdot 6 \cdot 5 \cdot 4}{6 \cdot 4} \times \frac{5 \cdot 4 \cdot 3}{2 \cdot 3}$$

$$b) \binom{7}{4} \binom{5}{1} = 35 \times 5 = 175$$

$$c) \binom{7}{4} \binom{5}{1} + \binom{7}{5} \binom{5}{0}$$

$$= 35 \times 5 + 21 \times 1 = 196$$

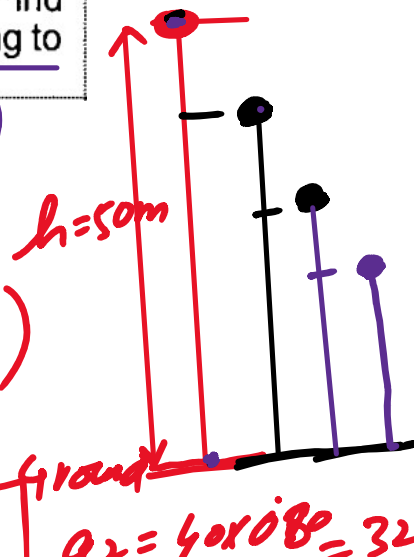
$$d) \binom{5}{0} \binom{7}{5} + \binom{5}{1} \binom{7}{4} = 1 \times 21 + 5 \times 35 = 196$$

A bouncing ball rebounds 80% of its previous height. If the ball is dropped from a height of 50 meters. Find the total distance traveled by the ball before coming to rest.

Solution: $S = H \left(\frac{1+r}{1-r} \right) = 50 \left(\frac{1+0.8}{1-0.8} \right)$

Total distance = $50m + 2(\text{Rebound distance})$

$$S_{\infty} = 50 + \frac{2a}{1-r} \rightarrow 196$$



$$S_{\infty} = S_0 + \frac{24}{1-0.8} \rightarrow (2)$$

Given, $q = 80\% = 0.8$ $q_1 = 50 \times 0.8 \Rightarrow q_1 = 40$ $q_2 = 40 \times 0.8 = 32$
 $40 + 32 + 25.6 + \dots$

How many words can be formed by using the letters from the word "EDUCATION" such that all the vowels are never together?

$$(2 \Rightarrow) S_{\infty} = 50 + 2 \times \frac{40}{1-0.8} = 450m$$

Hence, total distance = 450m

Verify that: $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$ $(n-1)!(n+1)!$

$$L.H.S = \binom{n}{r} + \binom{n}{r-1} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!}$$

$$= \frac{n!}{r(r-1)!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)(n-r)!}$$

$$= \frac{n!}{(r-1)!(n-r)!} \left[\frac{1}{r} + \frac{1}{n-r+1} \right]$$

$$= \frac{n!}{(r-1)!(n-r)!} \left[\frac{n-r+1 + r}{r(n-r+1)} \right] = \frac{(n+1)n!}{r(r-1)!(n-r+1)!}$$

$$= \frac{(n+1)!}{r!(n-r+1)!} = \binom{n+1}{r} = R.H.S$$

Verify that: ${}^nP = {}^{n-1}P + {}^{n-1}P$

Verify that: ${}^nP_r = {}^{n-1}P_r + r \cdot {}^{n-1}P_{r-1}$

$$LHS = {}^nP_r + r \cdot {}^{n-1}P_{r-1} = \frac{(n-1)!}{(n-r-1)!} + r \cdot \frac{(n-1)!}{(n-r)!}$$

$$= \frac{(n-1)!}{(n-r-1)!} + \frac{r \cdot (n-1)!}{(n-r)!}$$

$$= \frac{(n-1)!}{(n-r-1)!} + \frac{r \cdot (n-1)!}{(n-r)(n-r-1)!}$$

$$= \frac{(n-1)!}{(n-r-1)!} \left[1 + \frac{r}{n-r} \right] = \frac{(n-1)!}{(n-r-1)!} \left[\frac{n-r+r}{n-r} \right]$$

A harmonic sequence has 3rd term $\frac{1}{8}$ and 5th term

$\frac{1}{20}$. Find:

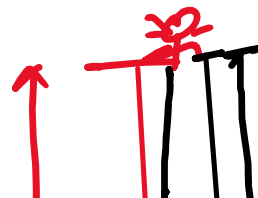
- (a) the general term and
(b) the 15th term of the sequence.

$$= \frac{n(n-1)!}{(n-r)(n-r-1)!}$$

$$= \frac{n!}{(n-r)!} = {}^nP_r = RHS$$

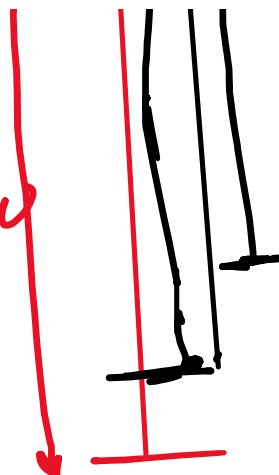
$\Rightarrow LHS = RHS$
Hence proved

A falcon dives from a height of 100 meters to catch its prey and then ascends back to 85% of its previous height after each unsuccessful attempt. Find the total vertical distance travelled by the falcon



height after each unsuccessful attempt.
Find the total vertical distance travelled by the falcon before stopping its attempts.

Total Distance
 $= 100 + 2 \left(\text{Sum of Rebound Heights} \right)$



$r = 0.85$ ✓

Sum of Rebound Heights $= S_{\infty} = \frac{a}{1-r}$

G.P. $(100)(0.85), (100)(0.85)^2, \dots$
 $85, 72.25, \dots \rightarrow \boxed{a = 85m}$

(1) total distance $= 100 + 2 \times \frac{85}{1-0.85} = 1233.33m$

A bakery offers 2 types of breads, 7 types of fillings and 5 types of sauces for a custom sandwich:

- (a) How many different sandwich combinations can be made? ✓
 (b) How many weeks can a customer eat a different sandwich each day without repetition?

Sol ~ Breads: 2 choices

Fillings = 7 choices

Sauces = 5 choices

a) No. of different sandwiches $= \binom{2}{1} \binom{7}{1} \binom{5}{1}$

b) Weeks: $\frac{70}{7} = 10 \text{ weeks}$ $= 2 \times 7 \times 5 = 70$

For an arithmetic-geometric series:

(1) (1) (1) (1)

For an arithmetic-geometric series:

$$1(1) + 3\left(\frac{1}{2}\right) + 5\left(\frac{1}{4}\right) + 7\left(\frac{1}{8}\right) + 9\left(\frac{1}{16}\right) + \dots$$

- (a) Find n^{th} term of the series ,
 (b) Sum the series up to n terms and ✓
 (c) Sum the series up to infinite terms ✓

$$T_n = \{a + (n-1)d\} r^{n-1} \rightarrow \textcircled{1}$$

AP: $1 \rightarrow 3, 5, 7, \dots$
 $\boxed{a=1}, \boxed{d=2}$

G.P.
 $1, \frac{1}{2}, \left(\frac{1}{2}\right)^2, \dots$
 $\boxed{r = \frac{1}{2}}$

$$T_n = \left[1 + (n-1)2\right] \left(\frac{1}{2}\right)^{n-1}$$

$$T_n = [1 + 2n - 2] \left(\frac{1}{2}\right)^{n-1}$$

$$\boxed{T_n = (2n-1) \left(\frac{1}{2}\right)^{n-1}}$$

$$\textcircled{31\frac{1}{2}}$$

A basketball club has 7 forwards and 5 guards. In how many ways a team of 5 can be selected if it includes:

- (a) 3 forwards and 2 guards?
 (b) 4 forwards and 1 guard?
 (c) At least 4 forwards?
 (d) At the most 1 guard?

$$b) S_n = \frac{a}{1-r} + \frac{dr(1-r^n)(1-r)}{1-r^2}$$

$$c) S_{\infty} = \frac{a}{1-r} + \frac{dr}{1-r^2}$$

$$a) \binom{7}{3} \binom{5}{2} \checkmark$$

$$c) \binom{7}{5} + \binom{7}{4}$$

a) $\binom{3}{1} \binom{2}{1}$ c) $\binom{7}{4} \binom{5}{1} + \binom{7}{5} \binom{5}{0}$
 b) $\binom{7}{4} \binom{5}{1}$ ✓ d) $\binom{5}{0} \binom{7}{3} + \binom{5}{1} \binom{7}{4}$ ✓

A team of 3 students is to be selected from 7 boys and 5 girls.

- a. In how many ways the team has 2 boys and 1 girl?
 b. In how many ways the team has 1 boy and 2 girls?

sol a) $\binom{7}{2} \binom{5}{1}$ ✓
 b) $\binom{7}{1} \binom{5}{2}$ ✓

Five brothers sat in a row for a group photo. Where two of them were twins. Find the number of arrangements when:

- a. Twins were treated distinct. ✓
 b. Twins stand together (not distinct) → *one diff*

a) $5! = 120$ arrangements
 $5! = 5 \times 4 \times 3 \times 2 \times 1$

b) $4 \times 3 \times 2 \times 1$
 $1! = 1$ way - 24 ways

$n! = nP_n$
 $5P_5 = \frac{5!}{0!}$
 $n! = nP_n$
 L R
 R H

$$= 4! \text{ way} = 24 \text{ ways}$$



Consider the letters of the word MANGO.

- Find number of ways to arrange all the letters.
- How many of these arrangements begin with M?

a) $5! = \frac{5}{1} \times \frac{4}{1} \times \frac{3}{1} \times \frac{2}{1} \times \frac{1}{1}$
 $= 120 \text{ ways}$

b) $\frac{M}{1} \times \frac{A, N, O}{4} \times \frac{3}{1} \times \frac{2}{1} \times \frac{1}{1}$
 $= 4! = 24 \text{ ways}$



if $|K| \geq 1$

-
- A hand-drawn diagram in red ink. It features a horizontal line with arrows at both ends. Two vertical tick marks are drawn perpendicular to the horizontal line, one on each side of the center. Below the horizontal line, there are two short vertical dashes, one under each tick mark.

b) $S_n = \frac{a(r^n - 1)}{r - 1}, |r| > 1$ 571
-5 < -1
 $|+r| > 1$

$$\begin{array}{|l|l|} \hline \underline{+x > 1} & \\ \hline x > 1 & | -x > 1 \\ \hline & \textcircled{x < -1} \\ \hline \end{array}$$

- Find the common difference
- Find the sum of first 10 terms.

Find a Harmonic sequence whose 4th term is $\frac{1}{3}$ and 9th term is $\frac{1}{13}$.

Find values of n and r , if:

a. $\binom{n}{2} = 28$ b. $\binom{10}{r} = 120$

8. Find n so that $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ may be the harmonic means between a and b .

Handwritten solution for problem 8:

$$\frac{2ab}{a+b} = \frac{a^{n+1} + b^{n+1}}{a^n + b^n}$$

Handwritten notes:

- A red checkmark is next to the first fraction.
- A red circle is around the expression $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$.
- A red circle is around the expression $n = ?$.
- Below the red circle, the expression $a^{\frac{1}{n}} + b^{\frac{1}{n}}$ is written with a red checkmark.

$$\textcircled{n=1}$$

$$\frac{1+1}{\frac{1}{a} + \frac{1}{b}} =$$

$$\frac{2}{\frac{a+b}{ab}}$$

$$\frac{a^{\frac{1}{2}} + b^{\frac{1}{2}}}{a^{\frac{1}{2}}}$$

6. If the product of three numbers in G.P be 216 and their sum be 19, then find the numbers.

$$\rightarrow \left[\frac{a}{r}, a, ar \right]$$

$$\frac{a}{r} \times a \times ar = 216 \Rightarrow a^3 = 216$$

$$a = 6$$

✓
5. Evaluate $3 + 33 + 333 + \dots$ up to n terms.

$$a, 3a, 9a, 27a, \dots \Rightarrow (24)$$

~~$$a, 2d, a+d, a+2d, \dots$$~~

$$a, a+d, \dots$$

$$a+3d = 9-d \Rightarrow (24)$$

2. The sum of four numbers in A.P. is 24 and their product is 945. Find the numbers.

3. Find four numbers in A.P., whose sum is 6 and sum of whose square is 14.

14. Sara pays her first installment Rs. 8000 to insurance company for the vehicle. Each installment will increase by 5%. What total amount she will pay in 24 installments?

1. A rocket rises 20 feet in the first second, 60 feet in the 2nd second and 100 feet in the third second. If it continues at this rate, how many feet will it rise in the 20th second?

3. Majid retired after 30 years of employment. If his salary was Rs. 4500 in the first year and he received an increment of Rs. 820 at the end of each year of service. What was his total salary after 30 years?

5. A vacuum pump removes $\frac{1}{5}$ of the air from a sealed container on each stroke of its piston. What percent of the air remains after five strokes of the piston?

100% = $\frac{100}{100}$ (1)

$\frac{1}{5} \times 1 = \frac{1}{5}$

Case 1

$1, \frac{4}{5}, \left(\frac{4}{5}\right)^2, \left(\frac{4}{5}\right)^3, \left(\frac{4}{5}\right)^4, \left(\frac{4}{5}\right)^5$

a_1, a_2

$n = 6, r = \frac{4}{5}$

$a_n = ar^{n-1}$

$a_6 = 1 \left(\frac{4}{5}\right)^5$

Proof

$\frac{4}{5} \times \frac{4}{5} \times \frac{4}{5} \times \frac{4}{5} \times \frac{4}{5} = \frac{4^5}{5^5} = \frac{1024}{3125}$

$1 - \frac{1}{5} = \frac{4}{5}$

$\frac{4}{5} \div \frac{4}{5} = 1$

$\frac{4}{5} \div \frac{4}{5} = 1$

$\frac{4}{5} \div \frac{4}{5} = 1$

$\frac{4}{5} \div \frac{4}{5} = 1$

$$a_6 = 1 \left(\frac{4}{5}\right)^5 \Rightarrow$$

$$a_5 = a_1 \left(\frac{4}{5}\right)^4 = \frac{4}{5}, \left(\frac{4}{5}\right)^2, \left(\frac{4}{5}\right)^3, \dots, \left(\frac{4}{5}\right)^5$$

26. A Ping-Pong ball is dropped from a height of 16 ft and always rebounds one-fourth of the distance fallen. How high does it rebound the 6th time?

$$a_5 = \left(\frac{4}{5}\right)^5 \times 100' = 32.8'$$

30. A 5-day rain caused the river to rise. After the first day, the river rose one inch. Each day the rise in the river tripled. How much had the river risen after 5 days?

27. A city has a current population of 100,000 and the population is increasing by 3% each year. What will the population be in 15th years?

$a_1 = 100,000$

$r = 1 + 3\% = 1.03$

$n = 15$ $a_{15} = ?$

$a_{15} = 100,000 (1.03)^{15}$

5%
 $100,000$
 $a_n = a_1 r^{n-1}$

16. A hot-air balloon rises 80ft in the first minute of flight. If in each succeeding minutes the balloon rises only 90% as far as in the previous minute, what will be its maximum altitude if it is allowed to rise without limit?

Example 37: A new virus is on a remote area. On day one, there were 10 people infected, with the number of new infections increasing at a rate of 40% per day.

- Find the expected number of infected people on the 7th day.
- Find the expected number of infected people during week (7 days).

A man deposits in a bank Rs. 2000 in the first year, Rs. 4000 in the second year, Rs. 8000 in the third year and so on. Find the amount he will have deposited in the bank by the fifth year.

4. You save Rs. 1 in the first day. Then each day thereafter, save double the amount you saved the day before. Find the amount you should save in the 20th day of your plan.

9. The number of bacteria in a culture increased geometrically from 16000 to 1215000 in 5 days. Find the daily rate of increase assuming the rate to be constant.

$$\begin{aligned}
 a_1 &= 16000 \\
 a_5 &= 1215000 \\
 a_n &= a r^{n-1} \\
 a_5 &= a r^4 \Rightarrow r^4 = \frac{1215000}{16000}
 \end{aligned}$$

$$a_5 = 9 \times 7 \Rightarrow r' = \frac{1}{16000}$$

$$r = \left(\frac{1215 \text{ mm}}{16000} \right)^{\frac{1}{4}}$$

Example 6: How many different arrangements of the letter used in the word EVENING can be made by using all the letters?

Example 8: There are 5 men and 5 women in a party. Find the number of ways in which they can be seated at a round table if:

- (i) anyone can occupy any seat,
- (ii) men and women have alternate seats.

13. In an examination, a candidate has to pass in each of 6 subjects. In how many ways he cannot qualify the examination?

8. There are 10 points on a circle. Find the number of (i) lines (ii) triangles that can be drawn?

6. How many 4-digit numbers can be formed with the digits 1, 2, 3, 4, 5, 6 when the repetition of the digits is allowed?

$n = 6$ digits

$$\begin{array}{ccccccc} \underline{6} & \times & \underline{6} & \times & \underline{6} & \times & \underline{6} \\ \uparrow & & & & & & \\ \textcircled{1, 2, 3, 4, 5, 6} & & & & & & \end{array}$$

$$N = \underline{\hspace{2cm}} = 6^4 \checkmark$$

How many words can be formed from the letters of the word FRIDAY? How many of them will end with F?

How many 4-digit odd numbers can be formed using the digits 1, 2, 3, 4 and 5 if no digit is repeated.

Odd nos. = 1, 3, 5

Total
ways = $\frac{4}{\text{---}} \times \frac{3}{\text{---}} \times \frac{2}{\text{---}} \times \frac{3}{\text{---}}$
1, 3, 5
 = 72 numbers

10. In how many ways a group of 10 girls can be divided into two groups of 3 and 7 girls.

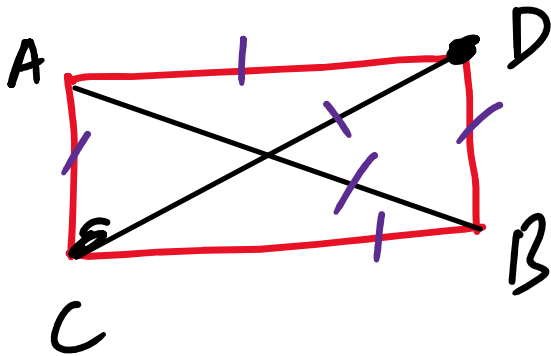
Out of 5 men and 3 women, a committee of 3 is to be formed. In how many ways can it be formed if at least one man is selected?

$$n=10$$



$$\binom{10}{3} \checkmark \quad n \checkmark \quad \binom{n}{3} \checkmark$$

Number of diagonals in n -sided polygon is 35. Find the number n ?



$n = \underline{4\text{-sided fig}}$

$$\# \text{ of lines} = \binom{n}{2} = \binom{4}{2} = 6$$

$$\# \text{ of diagonals} = 6 - n = 6 - 4 = 2$$

$$\boxed{\# \text{ of diagonal} = \binom{n}{2} - n}$$

$$\binom{n}{2} - n = 35 \Rightarrow \frac{n!}{2!(n-2)!} - n = 35$$

$$\frac{n(n-1)(n-2)!}{2(n-2)!} - n = 35 \Rightarrow \frac{n(n-1)}{2} - n = 35$$

In how many ways can letters of word MACHINE be arranged so that all the vowels are never together?



Find sum of the series

(a) $\sum_{i=1}^n \frac{i}{7^i}$ and (b) $\sum_{i=1}^{\infty} \frac{i}{7^i}$

a) $S_n = \sum_{i=1}^n \frac{i}{7^i} = \frac{1}{7^1} + \frac{2}{7^2} + \frac{3}{7^3} + \dots + n \text{ term}$

For A.P.: $1, 2, 3, \dots$ $a=1, d=1$

For G.P.: $\frac{1}{7^1}, \frac{1}{7^2}, \frac{1}{7^3}, \dots$

$r = \frac{1}{7}$ $\Rightarrow r = \frac{\frac{1}{7^2}}{\frac{1}{7^1}} = \frac{1}{7^2} \times \frac{7^1}{1}$

$S_n = \sum_{i=1}^n \frac{i}{7^i} = \frac{a}{1-r} + \frac{dr(1-r^n)}{(1-r)^2} - \frac{(a+nd)r^n}{1-r}$

$$S_n = \frac{1}{1 - \frac{1}{7}} + \frac{(1)(\frac{1}{7})(1 - (\frac{1}{7})^n)}{(\frac{1}{7})^2} - \frac{(1+n)(\frac{1}{7})^n}{1 - \frac{1}{7}}$$

$$b) S_{\infty} = \sum_{i=1}^{\infty} \frac{i}{7^i}$$

$$S'_{\infty} = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$$

$$= \frac{1}{1 - \frac{1}{7}} + \frac{\frac{1}{7}}{(\frac{1}{7})^2} = \checkmark$$

Given that $5, (5+3)2, (5+6)2^2, (5+9)2^3, \dots$

a. Show that given sequence is arithmetic geometric.

3+5

b. Find the sum of first n-terms of the sequence.

a) A.P.: $5, 5+3, 5+6, 5+9, \dots$

$5, 8, 11, 14, \dots$

$$a = 5, d = 8 - 5 = 3$$

$$\boxed{a=5}, \boxed{d=3}$$

For G.P $2^0, 2^1, 2^2, 2^3, \dots$

$$\boxed{r=2} \checkmark$$

$$b) S_n = \sum_{i=1}^n T_n \rightarrow \textcircled{1}$$

$$T_n = [a + (n-1)d] r^{n-1}$$

$$T_n = [5 + (n-1)3] 2^{n-1} \rightarrow 5 + 3n - 3$$

$$\boxed{T_n = [3n + 2] 2^{n-1}}$$

$$S_n = \sum_{i=1}^n (3i + 2) 2^{i-1}$$

$$S_n = \frac{a}{1-r} + \frac{dr(1-r^n)}{(1-r)^2} - \frac{(a+nd)r^n}{1-r}$$

How many words can be formed by using the letters from the word "EDUCATION" such that all the vowels are never together?

EDU EUD $\Rightarrow$ Principle of counting

A, B, BA $\Rightarrow$ 2! way $\left. \begin{array}{l} A \rightarrow B \\ B \rightarrow A \end{array} \right\}$

(ABC)

ABC

A < BC ①
A < CB ②

B < AC ③
B < CA ④

3!

$3 \times 2 = 6 \text{ ways}$

B → A]

C → AB ⑤
C → BA ⑥

A ③, ④
3 x 2 x 1
ABC BC

How many words can be formed by using the letters from the word "EDUCATION" such that all the vowels are never together?

Sol Total letters in
EDUCATION = 9

Total arrangement of
writing word "EDUCATION" = 9! ways

of vowels = 5

of consonants = 4

5
Vowels

We combine all vowels together to
make it one letter/object/group

∴ ways of arranging 5 letters = 5!

of ways of arranging 5 letters = $5!$ ways

As there are 5 vowels $\underline{5} \times \underline{4} \times \underline{3} \times \underline{2} \times \underline{1}$ So
Total ways of arranging vowels = $5!$ ways

Total Arrangements when vowels are placed together = $5! \times 5!$ ways

No. of ways when vowels are NOT together = $9! - 5! \times 5!$

$362880 - 14400$
Total ways when vowels are not together = $348,480$ ways

V C (V) C V C V C V

A harmonic sequence has 2nd term $\frac{1}{6}$ and 4th term $\frac{1}{12}$
Find general term of the sequence.

Given: 11. D ✓ n 1 a = 1

Given: In H.P. ✓ $a_2 = \frac{1}{6}$, $a_4 = \frac{1}{12}$

To find, a_n in H.P. ✓

Sol: In A.P. $a_2 = 6$, $a_4 = 12$

$$a_2 = a + d = 6$$

$$a_4 = a + 3d = 12$$

$$2d = 6 \Rightarrow d = 3$$

$$a + 3 = 6 \Rightarrow a = 3$$

$$a_n = a + (n-1)d$$

$$a_n = 3 + (n-1)3 = 3 + 3n - 3$$

$$a_n = 3n \text{ in A.P.}$$

Thus in H.P.

$$a_n = \frac{1}{3n}$$

✓